Albert Zhao

Education

Carnegie Mellon University **Sep 2026 – May 2028**
Master of Science in Information Networking, Advised by Prof. Tianqi Chen Pittsburgh, PA
- Tuition Scholarship | Concentration: Machine Learning System & AI Infrastructure

University of Rochester **Sep 2022 – May 2026**
Bachelor of Science in Computer Science & Bachelor of Science in Business Rochester, NY
- Computer Science Major GPA: 3.9/4.0

Professional Experience

Carnegie Mellon University, Catalyst Group **Apr 2026 – Current**
Research Assistant Intern, Advised by Prof. Tianqi Chen Pittsburgh, PA
- Working on LLM inference, GPU kernels, and Agent benchmarking

Cornell University, Computer Systems Laboratory **Apr 2026 – Current**
Research Assistant Intern, Advised by Prof. Mohamed S. Abdelfattah Ithaca, NY
- Working on efficient quantization for LLM inference

Tencent, Hunyuan LLM team **Jan 2026 – Apr 2026**
AI Infrastructure Engineer Intern Beijing, China
- Contributed the agent execution runtime to the evaluation/distillation infrastructure for Hy3 Preview model(295B MoE, A21B, 256K-token ctx), decoupling agent logic from heterogeneous execution backends
- Optimized a Nix-based runtime injection system that packages execution dependencies into one immutable Docker volume, enabling 2,500+ heterogeneous images to mount a shared read-only runtime and reducing environment complexity from $O(N \times M)$ to $O(N + M)$.
- Implemented a checkpoint/restore acceleration for runtime servers that leverages the immutable Nix store to reuse pre-warmed runtime snapshots across heterogeneous container images and cut service startup latency by 15+ seconds
- Migrated Socket.IO RPC to an asyncio/aiohttp + python-socketio implementation to prevent long-running task disconnects in Gongfeng Kubernetes sandboxes, sustaining stable operation with 100K+ concurrent workers

GetMyHome **Oct 2024 – Aug 2025**
Machine Learning Engineer Intern Los Angeles, CA
- Co-developed the core LLM multi-agent HomeGPT platform as a founding member, establishing it as an AI-powered real estate solution and directly securing $1.5M in pre-seed funding from Alumni Ventures and Wischoff Ventures
- Built Price-Estimate Agent in Apache Spark with 768-dim Sentence-BERT embedding time series and a BERT-LSTM model capturing temporal lag effects to generate house pricing reports that attracted 78K+ users to our platform
- Embedded scalable RAG infrastructure into the platform with FAISS vector indexing and LLM reranking, updating 20K+ daily property info with 70% improved relevance accuracy while maintaining $\leq 500ms$ response times

Haystack **May 2024 – Sep 2024**
Software Development Engineer Intern Los Angeles, CA
- Contributed an organization-wide universal search, delivering permissions-aware answers across every corner of the company's digital estate (Gmail, Microsoft 365, Workday, etc.)
- Designed dual-protocol API gateway serving 100K+ daily requests with Node.js/Express REST APIs that handle external integrations with 50+ third-party applications, and gRPC (Go/Gin) handles internal microservice
- Containerized search platform with Docker and Kubernetes (GKE) packaging Query, Index, and Auth services into lightweight containers that automatically scale from 10 to 100 instances during traffic spikes

Open-source & Competition Experience

FlashInfer Link ★ 5.6k 971 **Apr 2025 – Current**
- Contributed DeepSeek-V4 models support to the closed-loop, rigorous evaluation and production deployment of AI-agent-generated GPU kernels in FlashInfer-Bench, widely adopted by SGLang, vLLM, NVIDIA, Bosch, etc.

Kaggle Research Competition kaggle Link **Aug 2025 – Sep 2025**
- Achieved Silver Medal (Ranked 54th out of 1,282) in Kaggle Research competition for MakeDataCount by building VLLM-accelerated NLP pipeline with AWQ-quantized Qwen2.5-32B model on distributed GPU inference

Skills

Programming & Technology: Python, Go, Java, C, C++, JavaScript/TypeScript, FastAPI, Gin, Node.js, REST/JSON, gRPC/Protobuf, Kubernetes(GKE), LRU, Redis, CDN, OAuth, React, HTML/CSS, Git, FAISS
ML: Hugging Face, LangChain, PyTorch, TensorFlow, LLM/multi-agent/re-ranking, Retrieval Augmented, Deep Learning, Machine Learning, Computer Vision, Data Mining